Venkata Rohan Kambhampati

venkatarohank@gmail.com | Portfolio | LinkedIn | GitHub | +91-94911-11949

Professional Summary

AI Engineer specializing in **production-grade LLM systems** and **multi-agent orchestration**, with deep expertise in building scalable **Retrieval-Augmented Generation (RAG)** pipelines on Microsoft Azure. Delivered **70% latency reduction** (8.5s → **2.5s**) and **55% cost savings** ($12K → **$5.4K**) through vector search optimization, caching strategies, and efficient model orchestration. Designed high-precision **intent-based routing systems** achieving **95% classification accuracy** for **1,000+ enterprise users**. Strong experience across **retrieval quality**, **LLM evaluation**, **observability**, and **guardrails**, with proven ability to deploy end-to-end AI solutions from experimentation to **full-stack production deployment**, including React-based UIs, **API-driven backends** and cloud-native services.

Experience

Equinix Feb 2023 – Present

Senior Associate Software Engineer (Promoted from Intern) | *Python, FastAPI, Node.js, Azure OpenAI, LangChain, LangGraph*

- **Architected "Sidekick" Multi-Agent AI Platform:** Designed and deployed an enterprise LLM orchestration system using **LangGraph** with 4 specialized agents (HR, IT, Legal, General Query) serving 1,000+ employees; implemented intent classification using **GPT-4** with few-shot prompting achieving **95% routing accuracy** and 2s P95 latency through Redis caching & async execution.
- **Built Production RAG System (HRGPT):** Engineered end-to-end retrieval pipeline processing **50K+ ServiceNow documents**; implemented hybrid search combining dense vectors (text-embedding-3-large, 3072-dim) and BM25 with **Azure AI Search**, achieving **0.89 MRR@10** and reducing irrelevant responses from 35% to 4% through metadata filtering and semantic reranking.
- **Optimized RAG Performance & Quality:** Reduced query latency from 8.5s to **2.5s** (70% improvement) via multi-level caching (Redis L1, vector cache L2) and parallel retrieval; implemented query decomposition for multi-hop questions, improving answer completeness by 60% measured via GPT-4-as-judge evaluation framework.
- **Engineered Real-time Data Ingestion Pipeline:** Built incremental sync system processing 5K+ ServiceNow updates daily using **Azure Functions** and **Service Bus**; implemented change data capture with delta checksums, semantic chunking (500 tokens, 50 overlap), and batch upserts reducing ingestion time by 75% while maintaining 99.9% data freshness.
- **Implemented LLM Observability & Guardrails:** Deployed end-to-end tracing using **LangSmith** and Azure Application Insights tracking token usage, latency (P50/P95/P99), and error rates; built PII detection layer using regex + NER models preventing 200+ potential data leaks, and implemented RBAC-based context filtering ensuring users only access authorized documents.
- **Deployed Cost Optimization Strategies:** Reduced monthly OpenAI API costs by **55%** ($12K to $5.4K) through prompt compression, streaming responses, and intelligent model routing (GPT-3.5-turbo for simple queries, GPT-4 for complex); implemented response caching achieving 40% cache hit rate for repeated queries.
- **Automated ERP Workflows:** Developed Node.js microservices integrating **SAP FieldGlass**, **Coupa**, and **MS Teams** enabling inline approvals via Adaptive Cards; automated 15K+ monthly supplier onboarding tasks reducing manual effort by 75% and approval cycle time from 5 days to 6 hours through webhook-driven workflows.

Projects

Distributed Stock Exchange Simulation : (GitHub) | *TypeScript, Node.js, React.js, RabbitMQ, PostgreSQL, Prisma, Docker*

- Crafted a **full-stack application** simulating **real-time stock trading**, containerized using **Docker** for efficient deployments.
- Engineered a **4-microservice architecture** using **Node.js**, **RabbitMQ**, and **PostgreSQL** for **asynchronous order processing**; leveraged **Prisma ORM**, enabled **real-time updates** via **WebSockets**, and used **Nginx** for **load balancing**.
- Implemented **message durability and failure handling** with RabbitMQ acknowledgements, retries, and idempotent consumers to ensure **reliable order processing** across distributed services.

Open Source Contributions

LightDash (PR #12119): Engineered threshold-based alerting mechanisms, reducing manual monitoring overhead by **30%**.
Rocket.Chat (PR #32475) : Developed the "Notify on Reactions" feature, leading to a **20% increase** in user engagement metrics.
Cal.com (PR #18192): Refactored booking flow logic to preserve custom question sequencing and data integrity.

Technical Skills

AI & LLM: Azure OpenAI, LangChain, LangGraph, Vector Databases, Azure AI Search, Prompt Engineering, LLM Observability.
System Design & Architecture: LLD, SOLID, Microservices, Event-Driven Architecture, CQRS, CAP Theorem, Distributed Systems
Programming Languages: Java, Python, JavaScript, TypeScript, SQL
Backend & Databases: FastAPI, Node.js, Express.js, Redis, Kafka, RabbitMQ, PostgreSQL, MySQL, MongoDB, Azure Cosmos DB
Frontend: React.js, Next.js, HTML, CSS, Tailwind CSS
Cloud & DevOps: Azure Function App, Azure App Service, Docker, NGINX, Git, GitHub Actions, CI/CD

Education

B.Tech in Computer Science and Engineering CGPA: 8.5
Mahindra University, Hyderabad July 2019 – May 2023

Achievements

Competitive Programming: Solved **1,100+ problems** across LeetCode, GFG, and InterviewBit.
Certifications: Certified in Python | Deep Learning | Node.js | React.js